

# Chapter 5

## The gauge field and electrostatics

### 5.1 The gauge field versus $\vec{E}$ and $\vec{B}$

The field strength is built from the gauge field as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (5.1)$$

For the mixed time-space components, this gives

$$-E^i = \partial_0 A_i - \partial_i A_0 \quad (5.2)$$

so that, with  $A_0 = -\phi$ ,

$$\vec{E} = \vec{\nabla} A_0 - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \equiv -\vec{\nabla} \phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \quad (5.3)$$

while the space-space components package the magnetic field

$$\vec{B} = \vec{\nabla} \times \vec{A} \quad (5.4)$$

Under the gauge symmetry

$$A_\mu \rightarrow A_\mu + \partial_\mu \Lambda \quad (5.5)$$

the potentials shift as

$$\phi \rightarrow \phi - \frac{1}{c} \frac{\partial \Lambda}{\partial t} \quad , \quad \vec{A} \rightarrow \vec{A} + \vec{\nabla} \Lambda \quad (5.6)$$

Note that  $F_{\mu\nu}$  can be viewed as a matrix

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & B^3 & -B^2 \\ E^2 & -B^3 & 0 & B^1 \\ E^3 & B^2 & -B^1 & 0 \end{pmatrix} \quad (5.7)$$

## 5.2 Lorentz transformations of $\vec{E}$ and $\vec{B}$

The field strength is a rank  $(2, 0)$  tensor in its upper-index form, transforming as

$$F^{\mu'\nu'} = \Lambda^{\mu'}_{\mu} \Lambda^{\nu'}_{\nu} F^{\mu\nu} \quad (5.8)$$

Say  $\mathcal{O}$  and  $\mathcal{O}'$  have parallel axes, with  $\mathcal{O}'$  moving with velocity  $\vec{v}$  with respect to  $\mathcal{O}$ . One can then show

$$\begin{aligned} \vec{E}'_{\perp} &= \gamma \left( \vec{E}_{\perp} + \frac{\vec{v}}{c} \times \vec{B} \right) \quad , \quad \vec{E}'_{\parallel} = \vec{E}_{\parallel} \\ \vec{B}'_{\perp} &= \gamma \left( \vec{B}_{\perp} - \frac{\vec{v}}{c} \times \vec{E} \right) \quad , \quad \vec{B}'_{\parallel} = \vec{B}_{\parallel} \end{aligned} \quad (5.9)$$

– a Lorentz transformation mixes  $\vec{E}$  and  $\vec{B}$ . Similarly, the current transforms as a 4-vector

$$j^{\nu'} = \Lambda^{\nu'}_{\nu} j^{\nu} \quad (5.10)$$

(one can also show that  $Q = Q'$ , with  $Q \equiv \int j^0 dV$ ).

## 5.3 Boundary conditions

Consider a surface separating two regions, labeled 1 and 2, with unit normal  $\hat{n}$  pointing from 1 into 2, as shown in the Figure. Applying

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho \quad , \quad \vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t} \quad (5.11)$$

across the surface leads to

$$\left( \vec{E}_2 - \vec{E}_1 \right) \cdot \hat{n} = 4\pi\sigma \quad , \quad \left( \vec{E}_2 - \vec{E}_1 \right) \times \hat{n} = 0 \quad (5.12)$$

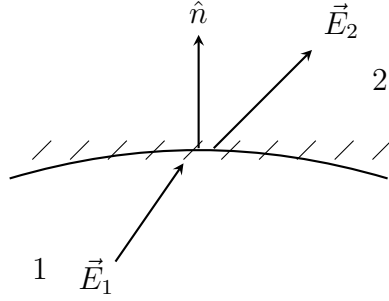


Figure 5.1: A surface separating regions 1 and 2, with unit normal  $\hat{n}$  and the electric field on either side.

Similarly

$$\left(\vec{B}_2 - \vec{B}_1\right) \cdot \hat{n} = 0 \quad , \quad \left(\vec{B}_2 - \vec{B}_1\right) \times \hat{n} = \frac{4\pi}{c} \vec{K} \quad (5.13)$$

where  $\vec{K}$  is the surface current density (current per length).

## 5.4 Electrostatics

### 5.4.1 The point particle

Consider a point particle at rest at the origin:  $y^0 = ct$ ,  $\vec{y}(t) = 0$ . Then

$$\rho = q \delta^3(\vec{x}) \quad , \quad \vec{j} = 0 \quad (5.14)$$

where  $\delta^3$  is the delta function, normalized as  $\int \delta^3(\vec{x}) dV = 1$  so that  $q = \int \rho dV$ . This gives

$$\vec{\nabla} \cdot \vec{j} = 0 \quad , \quad \dot{\rho} = 0 \quad (5.15)$$

– a static configuration. Maxwell's equations become

$$\vec{\nabla} \cdot \vec{E} = 4\pi q \delta^3(\vec{x}) \quad , \quad \vec{\nabla} \cdot \vec{B} = 0 \quad , \quad \vec{\nabla} \times \vec{E} = 0 \quad , \quad \vec{\nabla} \times \vec{B} = 0 \quad (5.16)$$

which are solved by

$$\vec{B} = 0 \quad , \quad \vec{E} = \frac{q \vec{x}}{x^3} \quad (5.17)$$

with  $x \equiv |\vec{x}|$ . To check this, we need to remember that  $\delta^3(\vec{x})$  is a distribution, defined only inside an integral.

### 5.4.2 An aside on distributions

The delta function is defined by

$$f(a) = \int_{-\infty}^{\infty} f(x) \delta(x-a) dx \quad , \quad \int_{0-\epsilon}^{0+\epsilon} \delta(x) dx = 1 \quad (5.18)$$

Integration by parts gives

$$\int_{-\infty}^{\infty} f(x) \delta'(x-a) dx = -f'(a) \quad (5.19)$$

For a delta function of a function,

$$\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{\left| \frac{df}{dx}(x_i) \right|} \quad \text{where } f(x_i) = 0 \quad (5.20)$$

(derived by Taylor expanding near each  $x_i$ ). A useful representation is

$$\lim_{\sigma \rightarrow 0} \frac{1}{\sqrt{\pi} \sigma} e^{-x^2/\sigma^2} = \delta(x) \quad (5.21)$$

In multiple dimensions,

$$\begin{aligned} \delta^D(x-a) &= \delta(x^0-a^0) \delta(x^1-a^1) \cdots \delta(x^{D-1}-a^{D-1}) \\ \delta^4(x-a) &= \delta(x^0-a^0) \delta^3(\vec{x}-\vec{a}) \end{aligned} \quad (5.22)$$

and

$$\delta^D(C(x-a)) = \frac{\delta^D(x-a)}{|\det C|} \quad , \quad \delta^D(f(x)) = \frac{\delta^D(x-a)}{|\det \partial_\mu f^\nu(a)|} \quad (5.23)$$

the latter for a single zero of  $f$  at  $x=a$ .

### 5.4.3 Checking the point-charge field

We now check the field (5.17) as a distribution acting on a test function  $f$

$$\begin{aligned} \vec{\nabla} \cdot \vec{E}[f] &\equiv \int (\vec{\nabla} \cdot \vec{E}) f dV = \int \partial_i E^i f dV = - \int E^i \partial_i f dV \\ &= \lim_{\epsilon \rightarrow 0} - \int_{r>\epsilon} \frac{q x^i}{x^3} \partial_i f dV = \lim_{\epsilon \rightarrow 0} - \int_{r>\epsilon} \partial_i \left( \frac{q x^i}{x^3} f \right) dV \end{aligned} \quad (5.24)$$

where the last step uses

$$\partial_i \frac{x^j}{x^3} = \frac{\delta_i^j}{x^3} - \frac{3}{x^5} x^i x^j \Rightarrow \partial_i \left( \frac{x^i}{x^3} \right) = \frac{3}{x^3} - \frac{3}{x^3} = 0 \quad (5.25)$$

By Gauss's law, with  $da_i$  the outward normal area element on the sphere  $r = \epsilon$ ,

$$\begin{aligned} \vec{\nabla} \cdot \vec{E}[f] &= \oint_{r=\epsilon} f \frac{q x^i}{x^3} da_i = f(0) \frac{q}{\epsilon^2} 4\pi \epsilon^2 \\ &= 4\pi q f(0) = 4\pi \int \rho f dV \Rightarrow \vec{\nabla} \cdot \vec{E} = 4\pi \rho \end{aligned} \quad (5.26)$$

as required. And  $\vec{\nabla} \times \vec{E} = 0$  as needed, since  $\partial_i E^j$  is symmetric in  $ij$ . Hence, we write

$$\vec{\nabla} \cdot \left( \frac{\vec{x}}{x^3} \right) = 4\pi \delta^3(\vec{x}) \quad (5.27)$$

Writing  $\vec{E} = -\vec{\nabla}\phi$  gives

$$\phi = \frac{q}{x} \quad (5.28)$$

But  $\vec{\nabla}^2 \phi = -4\pi \rho = -4\pi q \delta^3(\vec{x})$ , hence

$$\vec{\nabla}^2 \left( \frac{1}{x} \right) = -4\pi \delta^3(\vec{x}) \quad (5.29)$$

By superposition, for a general  $\rho$  we build the solution out of point charges

$$\vec{E}(\vec{x}) = \int \rho(\vec{x}') \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} dV' \quad , \quad \phi(\vec{x}) = \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' \quad (5.30)$$